Design & Digital Prototype Documentation

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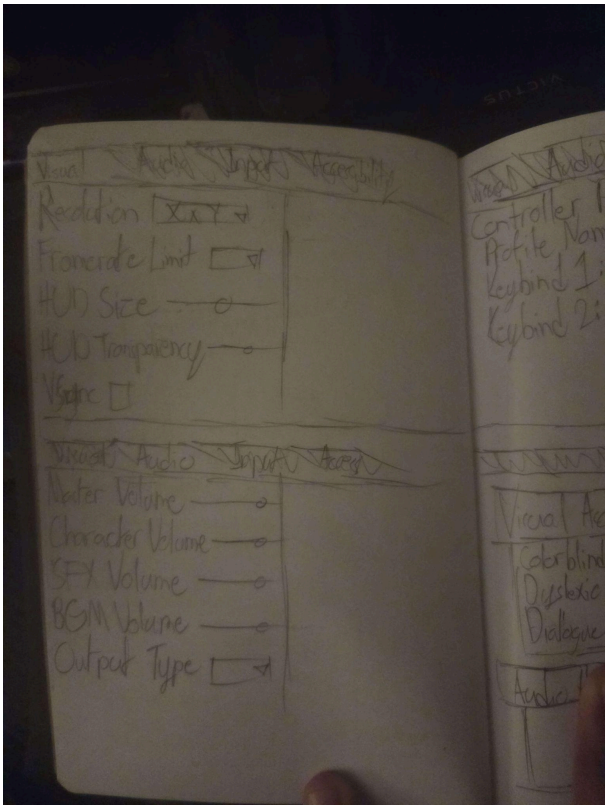
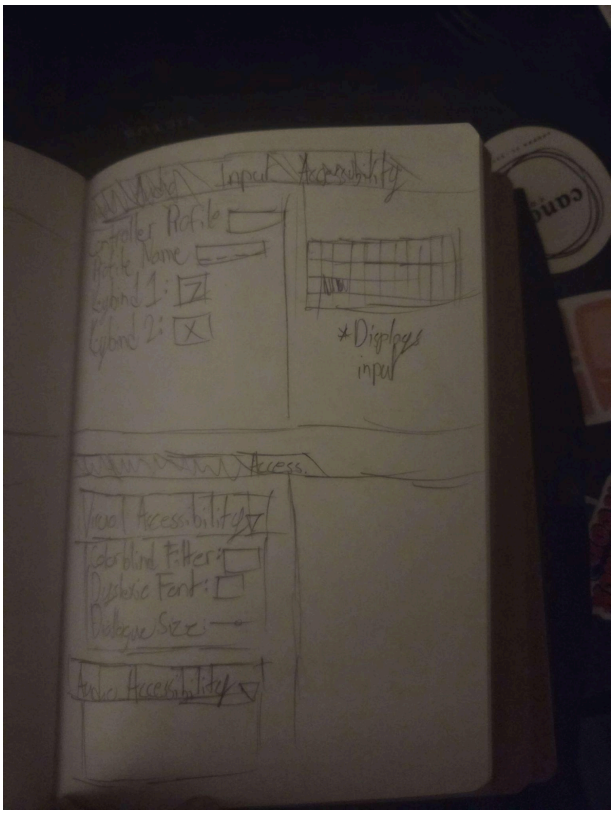
Instructions to Run Prototype:

Download the provided .exe and double-click on it to run.

Sketches:



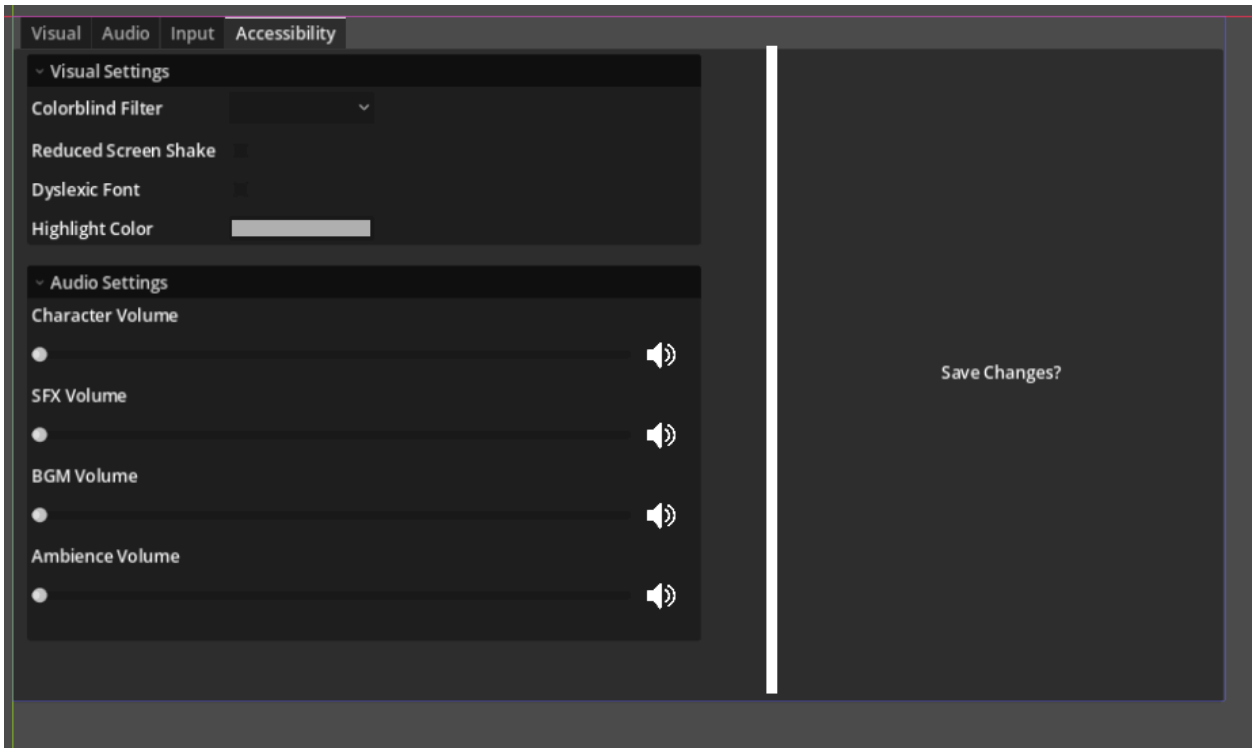


To begin, the first image on the left showcases our design for visual and audio options. We also have the names of the sub-menus right at the top. It's visible and to make sure that the user knows

what menu they're on, you can see how the scribble 'breaks' on whichever sub-menu the user is in. This showcases how we'll implement over 6 of our functional requirements. FR1 (Adjusting the sound of volume for specific sounds), FR2 (quick access to options), FR4 (Saving changed settings), FR5 (pausing the game allows users to go into settings), FR6 (adjusting the HUD), and FR7 (Clearly organized options). We have clearly labeled and organized options in their specific categories of settings. (Resolution in visual, output device in audio, etc). While it's not shown, our menu will maintain whatever options a user chose to change across game sessions. Video game option menus rely on this feature, as otherwise it would hinder the user's experience by having them change settings every single time they close the game and reopen it.

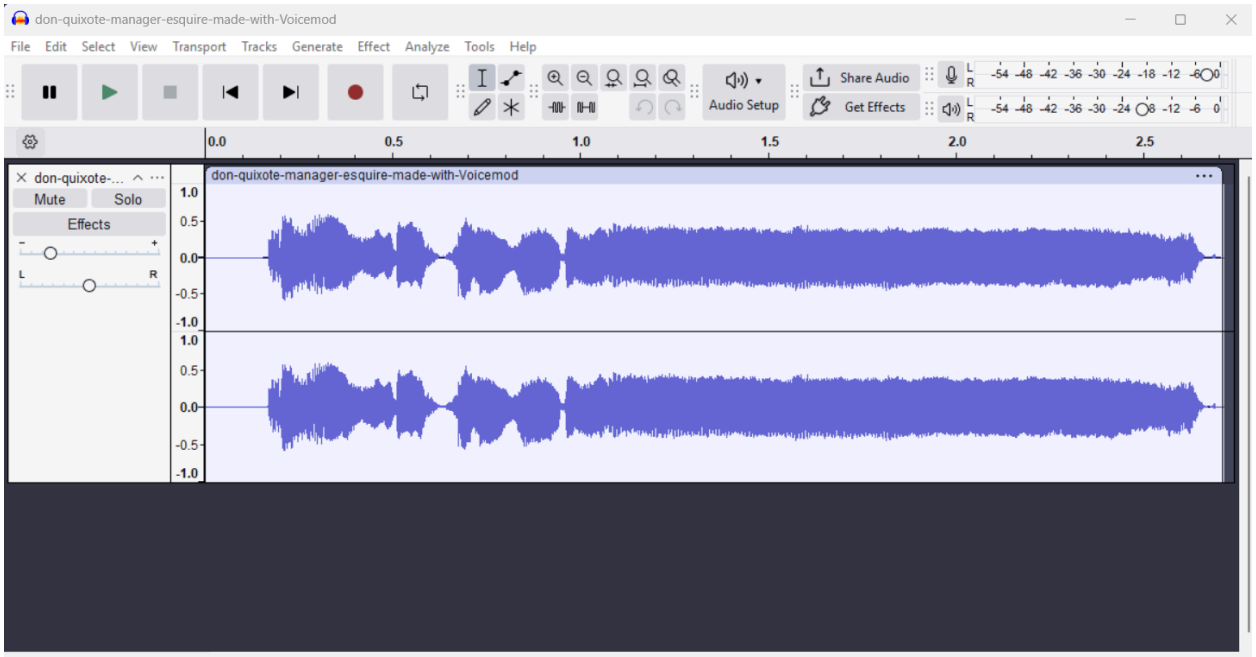
The other image shows the input and accessibility section of our menu. In input, we've shown a small window to the right that shows where the input they've just selected is on the keyboard. This won't be limited to just the input submenu, as we have that space reserved specifically for the window. For example, if the user were to navigate to the accessibility menu and change the colorblind filter to red-green mode, it would show how that filter changes the color of blood, foliage, and other examples of objects with the affected color. This would be extra important in roguelikes where there is no stopping within a run. A user can see what they're changing before jumping into things.

Implementations and Justifications:

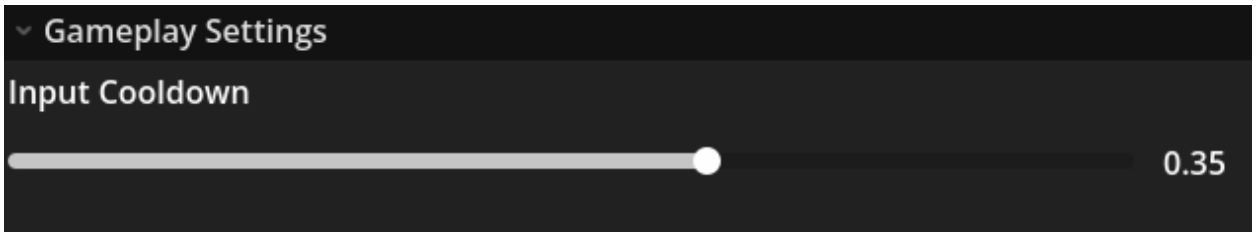


Here's a screenshot of the accessibility screen in our menu. We've put audio settings in here as unfortunately we've realized that to implement every other submenu within the prototype it would take much longer than the time we have currently and everything can be accomplished within the current tab at this phase anyway. We've implemented the accessibility tab since this is just a proof of concept.

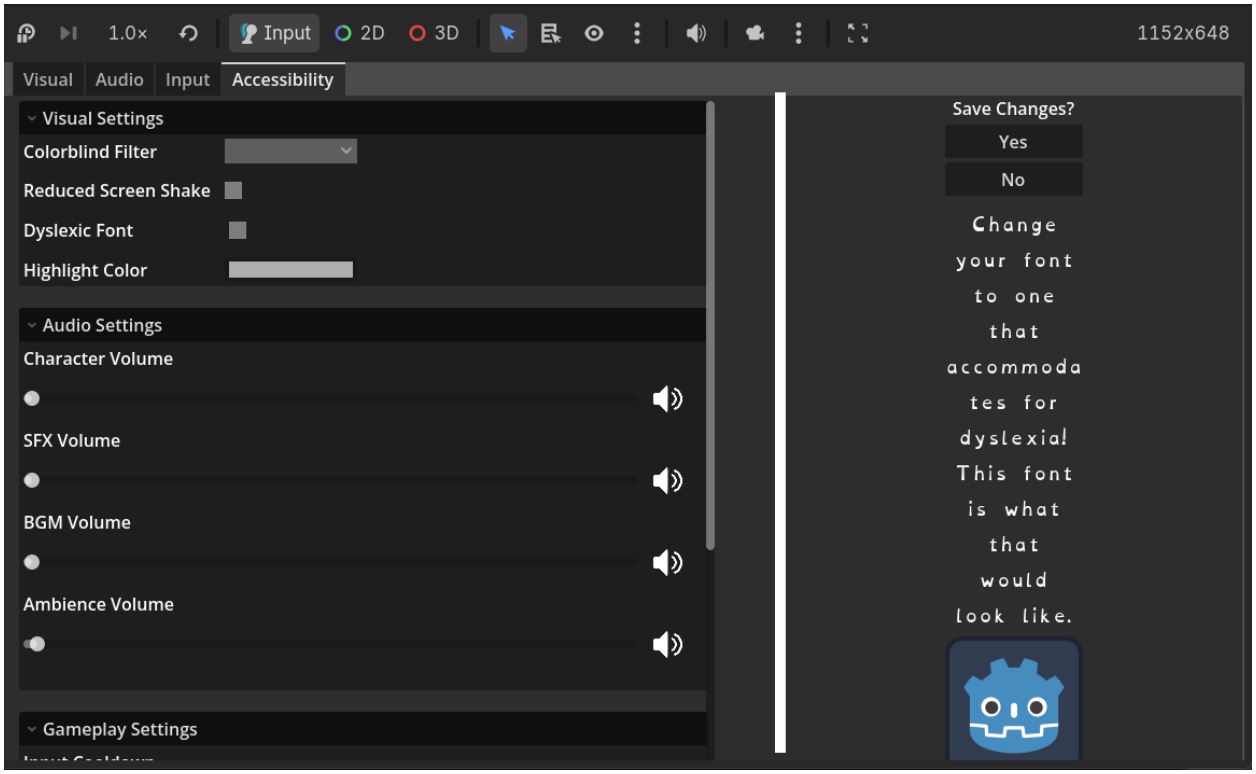
On the right, you'll see 'Save Changes?'. This is to show how we've implemented FR4, persistence of accessibility settings across sessions. This area will also house the option preview window, giving players a way to view what they're changing before they get back into playing. We also have some of our visual accessibility features to show. The colorblind filter, dyslexic font, reduced screen shake, etc.



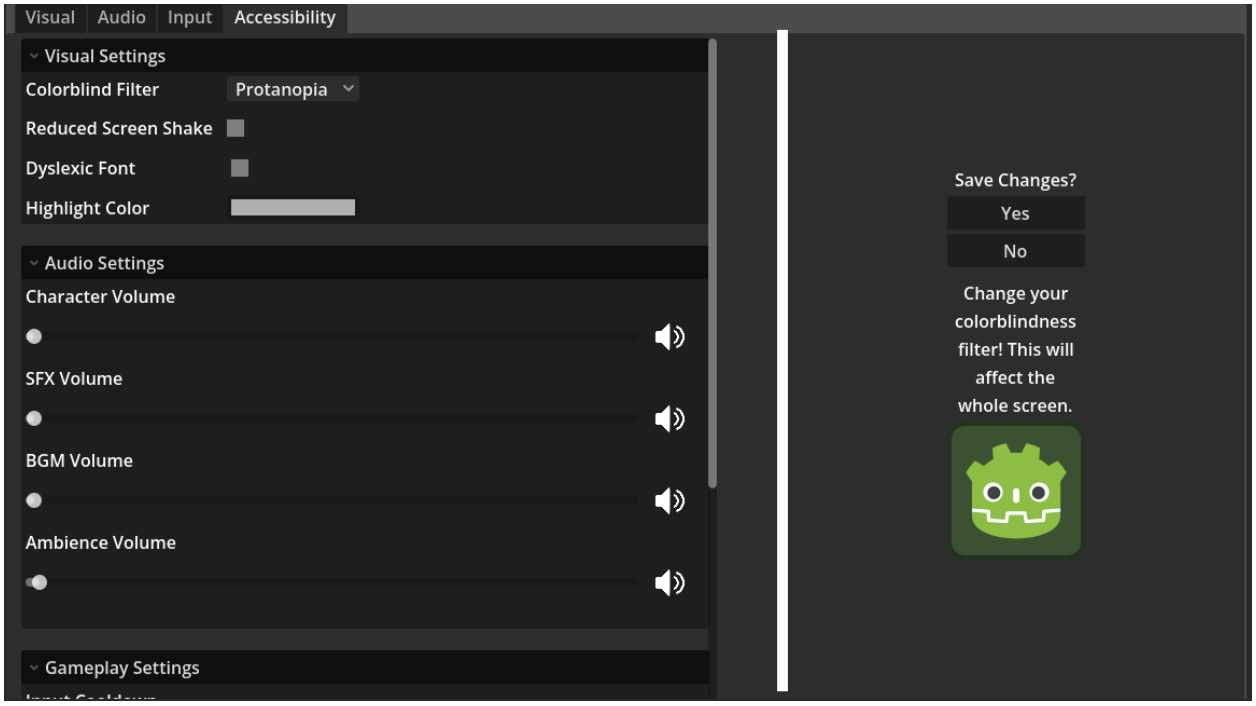
We've also added more feedback when you click buttons, a sound playing when you do so. This is Lukas in Audacity making sure that the audio file isn't so loud.



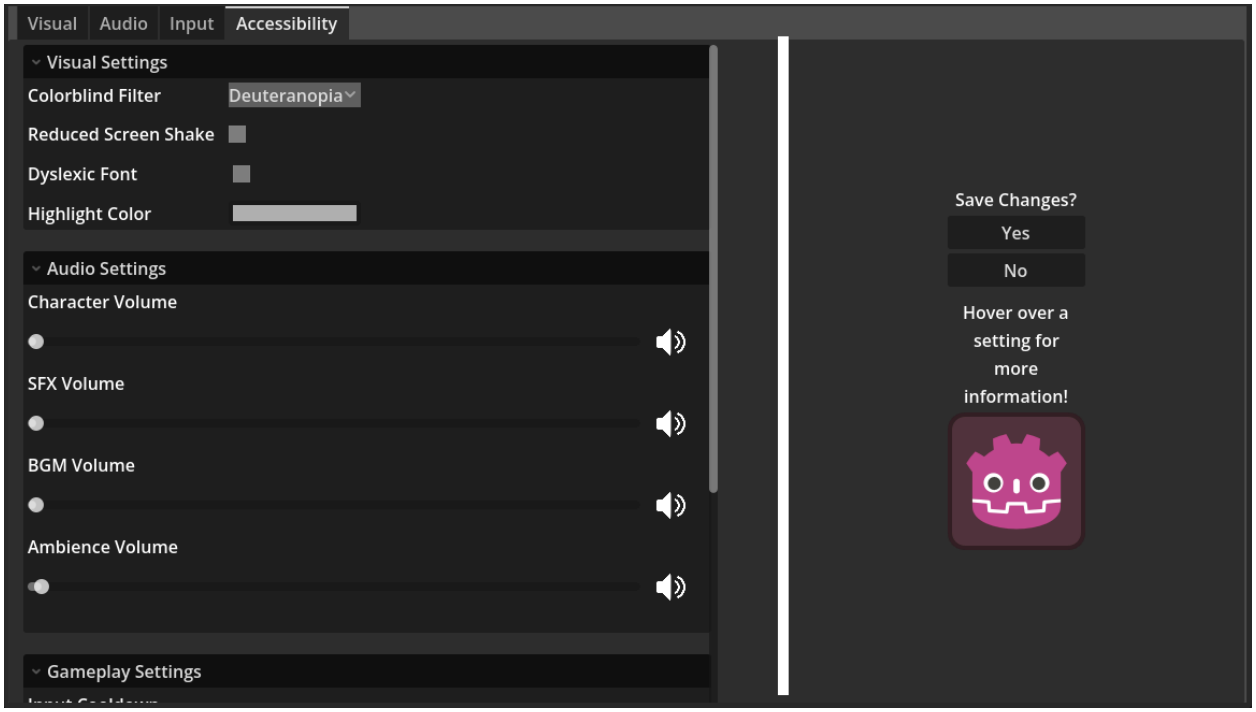
Here is another example of an accessibility option with granular controls. Input cooldowns help with those with diseases such as Parkinson's where they might accidentally press a button multiple times.



Here is a more polished version of the accessibility tab. Note that the preview window now shows the dyslexia font we've implemented along with the Godot video game engine's logo which we'll use to show off the colorblind filters. This screenshot shows the implementation of FR1, FR2, FR4, FR7, NFR1 (the menu is smooth to navigate) and NFR2 (which while hard to show, changes are snappy and quick to display on the right when applicable).



Here you'll see we've successfully implemented the colorblind filters. In this case, it's protanopia.



This one is Deuteranopia. You'll notice the changing of the logo along with the drop-list option labeled with the name of the type. These are the implementation of FR3, enabling colorblind filters along with a preview.

Contributions

Mason Adair	Documentation, Presentation Powerpoint
Jorge Canales	Documentation, Presentation Powerpoint
Lukas Jaspe	Prototyping Build
Sebastian Foster	Prototyping Build
Yannick Morsen	Presentation Powerpoint
Jack Zhang	Documentation, Presentation Powerpoint